

Recursive Offline Distillation via Epistemic Complexity Approximation

Anonymous ACL submission

Abstract

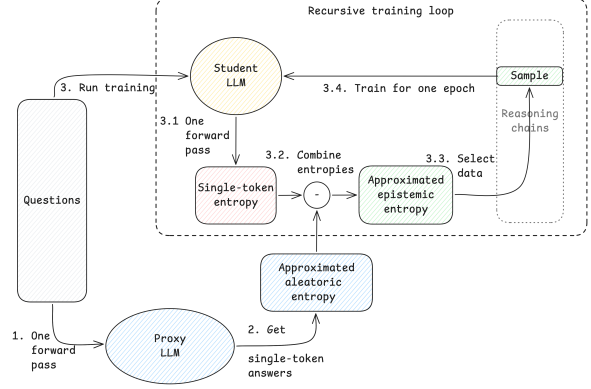


Figure 1: Recursive offline distillation scheme

Distillation of reasoning chains from Large Language Models (LLMs) into their smaller siblings is a standard training approach. Quite frequently researchers do not have full access to a teacher model and its internal representations, which limits the resulting performance. We propose a novel blueprint for efficient offline distillation that recursively uses lightweight epistemic and aleatoric complexity approximations for selective distillation. Specifically, for three open models (3-4B) at each epoch we estimate offline data complexity with a single-token answer entropy and use a medium model’s (70B) entropy as an aleatoric proxy. Next, we apply selective distillation only for a small subset of the reasoning chains with the highest corresponding epistemic component. Our approach consistently outperforms naive distillation on comparable computational budget (57.15 ± 1.01 vs 51.61 on average) and offers a much steeper learning curve (baseline final score on 15% of the data). Additionally, we present comparative analysis of methods to generate, standardize and augment synthetic reasoning chains when the teacher model does not perform perfectly on the target distillation domain in the first place. We publish our code and data to facilitate further research in this direction.

1 Introduction

2 Related Work

3 Methods

4 Results

4.1 Experimental Setup

4.2 Main Results

4.3 Synthetic Traces Generation

4.4 Ablation Study

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Limitations

A Example Appendix

This is an appendix.

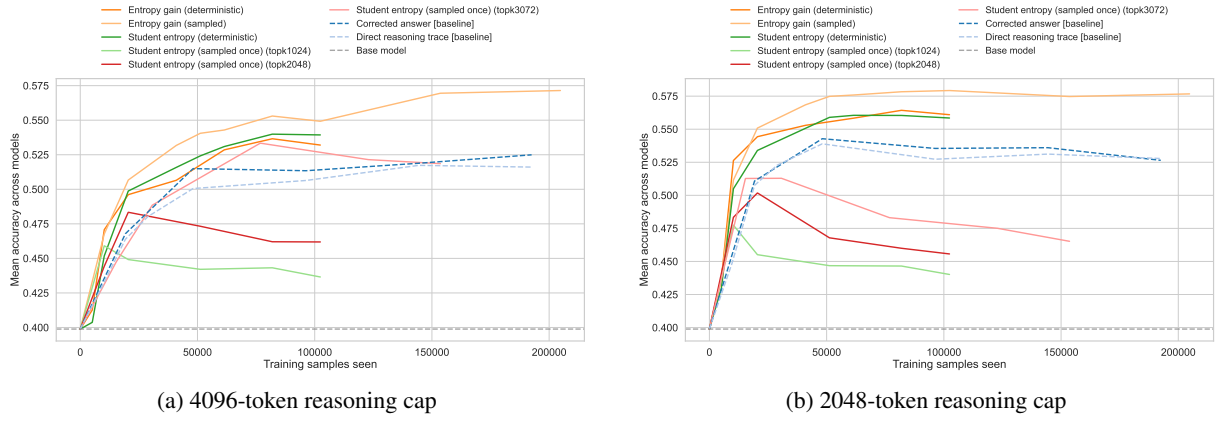


Figure 2: Distillation pipeline results across three models (Qwen-3B, LLaMA-3B, and Phi4-mini) on MMLU-Pro

Method	LLaMA-3B	Qwen-3B	Phi4-mini	Mean
Direct reasoning trace	46.09 / 45.55	51.45 / 50.37	57.27 / 62.43	51.61 / 52.78
Corrected answer	47.80 / 47.26	50.87 / 49.42	58.81 / 61.26	52.49 / 52.65
Student entropy (deterministic)	49.83 / 50.08	53.62 / 53.03	58.35 / 64.42	53.93 / 55.85
Entropy gain (deterministic)	<u>50.17 / 50.37</u>	51.04 / <u>53.28</u>	58.40 / 64.63	53.20 / <u>56.10</u>
Student entropy (sampled once, $k = 1024$)	41.35 / 40.11	39.44 / 39.07	50.17 / 52.87	43.65 / 44.01
Student entropy (sampled once, $k = 2048$)	42.19 / 41.06	43.18 / 43.02	53.20 / 52.62	46.19 / 45.57
Student entropy (sampled once, $k = 3072$)	42.02 / 42.81	56.77 / 40.86	56.77 / 55.90	51.86 / 46.52
	52.58 $\pm$ 1.94 /	56.64 $\pm$ 3.79 /	62.22 $\pm$ 2.42 /	57.15 $\pm$ 1.01 /
Entropy gain (sampled)	52.08 $\pm$ 2.28	56.86 $\pm$ 2.32	64.07 $\pm$ 4.52	57.67 $\pm$ 1.19

Table 1: Distillation pipeline last epoch accuracy (%) across three models (Qwen-3B, LLaMA-3B, and Phi4-mini) on MMLU-Pro, reported for 4096-token / 2,048-token thinking budget. Sampled entropy gain uses three seeds per model and includes $\pm$ 95% confidence-interval. Other methods are deterministic.

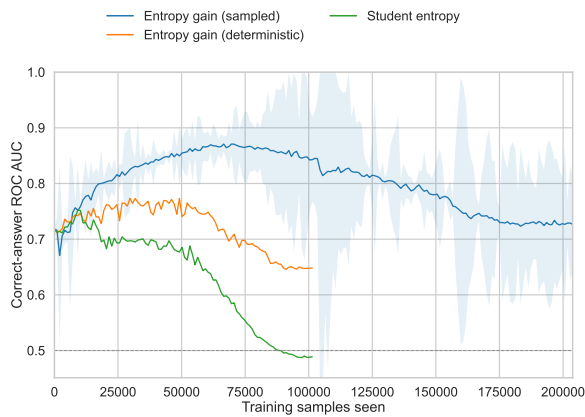


Figure 3: Student calibration

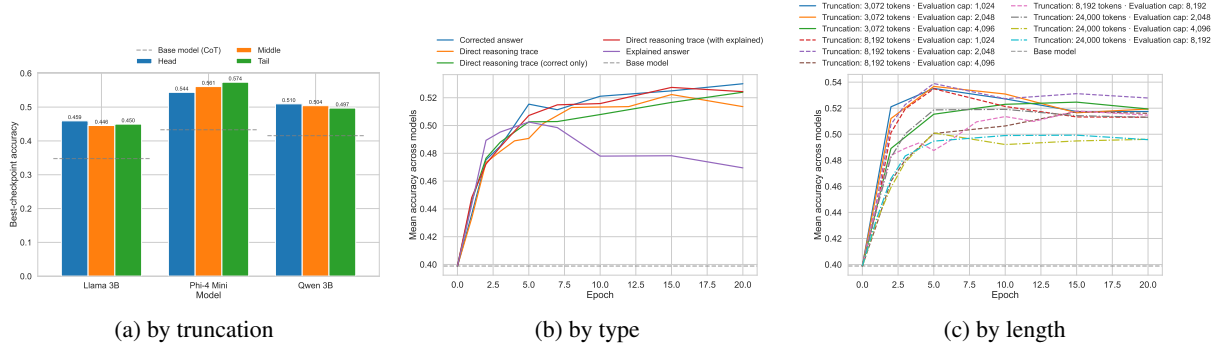


Figure 4: Comparison of synthetic traces

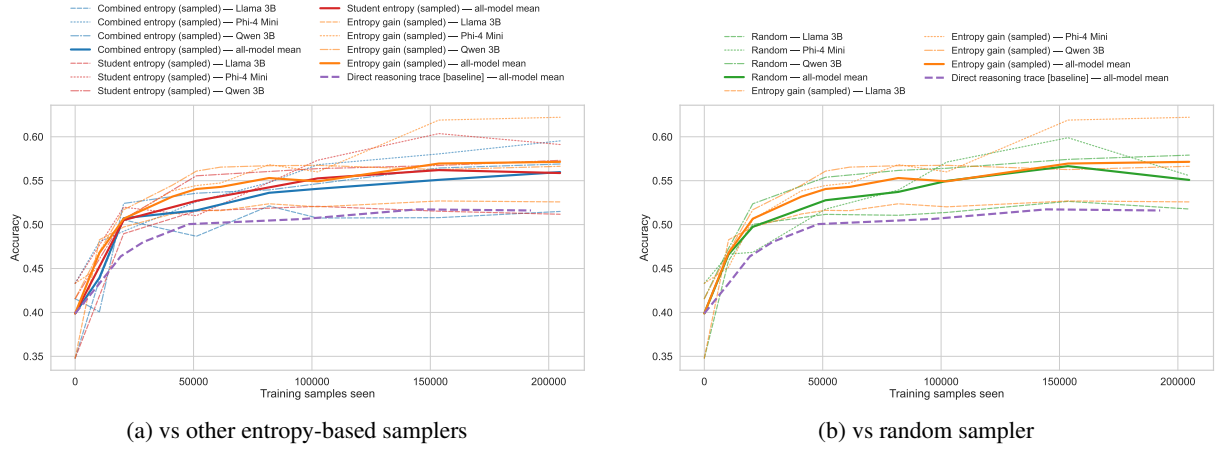


Figure 5: Comparison of entropy gain sampler

Sampler	LLaMA-3B	Qwen-3B	Phi4-mini	Mean
Random	51.77 ± 2.76	57.91 ± 1.40	55.57 ± 2.54	55.08 ± 0.80
Student entropy	51.18 ± 3.11	57.32 ± 1.52	59.12 ± 5.22	55.87 ± 1.40
Combined entropy	51.50	56.90	59.53 ± 2.11	55.98
Entropy gain	52.58 ± 1.94	56.64 ± 3.79	62.22 ± 2.42	57.15 ± 1.01

Table 2: Distillation pipeline last epoch accuracy for different sampling methods across three models (Qwen-3B, LLaMA-3B, and Phi4-mini) on MMLU-Pro, reported for 4096-token thinking budget. Each method is evaluated with 3 seeds per model and includes $\pm 95\%$ confidence-interval.